

Regenerative Cardiology Compendium

Clinical Studies of Cell Therapy in Cardiovascular Medicine Recent Developments and Future Directions

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Abstract: Given the rising prevalence of cardiovascular disease worldwide and the limited therapeutic options for severe heart failure, novel technologies that harness the regenerative capacity of the heart are sorely needed. The therapeutic use of stem cells has the potential to reverse myocardial injury and improve cardiac function, in contrast to most current medical therapies that only mitigate heart failure symptoms. Nearly 2 decades and >200 trials for cardiovascular disease have revealed that most cell types are safe; however, their efficacy remains controversial, limiting the transition of this therapy from investigation to practice. Lessons learned from these initial studies are driving the design of new clinical trials; higher fidelity of cell isolation techniques, standardization of conditions, more consistent use of state of the art measurement techniques, and assessment of multiple end points to garner insights into the efficacy of stem cells. Translation to clinical trials has almost outpaced our mechanistic understanding, and individual patient factors likely play a large role in stem cell efficacy. Therefore, careful analysis of dosing, delivery methods, and the ideal patient populations is necessary to translate cell therapy from research to practice. We are at a pivotal stage in the field in which information from many relatively small clinical trials must guide carefully executed efficacy trials. Larger efficacy trials are being launched to answer questions about older, first-generation stem cell therapeutics, while novel, second-generation products are being introduced into the clinical realm. This review critically examines the current state of clinical research on cell-based therapies for cardiovascular disease, highlighting the controversies in the field, improvements in clinical trial design, and the application of exciting new cell products. (*Circ Res.* 2018;123:266-287. DOI: 10.1161/CIRCRESAHA.118.311217.)

Key Words: cell- and tissue-based therapy ■ clinical trial ■ heart diseases ■ heart failure

Over the past 2 decades, there has been intense interest in harnessing and promoting the innate regenerative ability of injured organs to treat disease. Based on the longstanding research in stem cells for cardiovascular disease (CVD) and its translation into the clinical arena,¹⁻³ an approved product for clinical practice may be on the horizon. Fueled by exciting preclinical findings and growing expertise in cell isolation and handling, interest in cell-based therapy has remained steadfast despite inconsistent and sometimes disappointing clinical trial outcomes. From the first trials in humans⁴ to large, multicenter clinical trials,⁵⁻⁷ lessons from >200 clinical trials and ≈50 meta-analyses have shaped the understanding of how stem cells can improve cardiac function and have guided the search for the most efficacious approach. Despite the lack of a trial that conclusively demonstrates clinical efficacy, the safety and feasibility of several different cell types are now established, and larger phase III trials are underway that will address pivotal questions about efficacy. This field is at a stage in which important lessons have been learned that can help shape the design and execution of future clinical trials with novel, second-generation cell-based technologies.

Despite some disappointing clinical results and even negative trials, continued interest in stem cells for cardiac regenerative medicine (CRM) is spurred by a dire need for novel therapies for CVD.⁸ Ischemic heart disease is already the leading cause of death worldwide, accounting for ≈16% of all deaths and, as the global population ages, so will the prevalence of CVD.^{9,10} Surgical revascularization and percutaneous coronary intervention have improved survival after acute myocardial infarction (AMI), contributing to a rise in the prevalence heart failure (HF).¹¹ In the United States, HF is projected to increase from 6.5 million to >8 million cases by 2030.^{12,13} Fortunately, mortality from HF has fallen significantly in the past 2 decades,¹⁴ largely because of evidence-based therapies, such as angiotensin-converting enzyme inhibitors, β -blockers, mineralocorticoid receptor antagonists, and angiotensin receptor neprilysin inhibitors.¹⁵ Even with these medications, mortality from HF remains high and is extremely costly both in terms of medical expenses and disability-adjusted life years.¹⁰ Furthermore, drug therapy for HF only is efficacious in a subset of patients, and no therapy has been approved for HF with preserved ejection fraction or for worsening chronic HF.¹⁶ The

This manuscript was sent to Roger J. Hajjar, Consulting Editor, for review by expert referees, editorial decision, and final disposition.

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DOI: 10.1161/CIRCRESAHA.118.311217

Nonstandard Abbreviations and Acronyms

6MWD	6-minute walk distance
AMI	acute myocardial infarction
BMMNC	bone marrow mononuclear cell
BM-MSC	bone marrow–derived mesenchymal stem cell
CCTR	cardiovascular cell therapy research network
CDC	cardiosphere-derived cell
cMRI	cardiac magnetic resonance imaging
CRM	cardiac regenerative medicine
CSC	cardiac stem cell
CVD	cardiovascular disease
DVM	dysfunctional viable myocardium
ESC	embryonic stem cell
HF	heart failure
ICM	ischemic cardiomyopathy
iPSC	induced pluripotent stem cell
LV	left ventricle
LVEDV	left ventricular end-diastolic volume
LVEF	left ventricular ejection fraction
LVESV	left ventricular end-systolic volume
MI	myocardial infarction
MLHFQ	Minnesota Living with Heart Failure Questionnaire
MSC	mesenchymal stem cell
NIDCM	nonischemic dilated cardiomyopathy
NYHA	New York Heart Association
QoL	quality of life
TESI	transendocardial stem cell injection
TNF	tumor necrosis factor
UC-MSC	umbilical cord–derived mesenchymal stem cell

definitive treatment for end-stage HF is transplantation, and although because of limited availability of donor organs, the number of heart transplants performed per year in the United States has not changed during the past decade despite rising HF rates, making transplant an unrealistic option for most patients.¹⁷

The limit in treatments for CVD is because of the previously prevailing viewpoint that the heart is a terminally differentiated organ with little to no regenerative capacity. This paradigm shifted after the identification of a small resident population of multipotent cardiac stem cells (CSC), reframing the heart as an organ with the capacity for self-renewal.¹⁸ It is now understood that although they are terminally differentiated cells, there is a 1% to 2% cardiomyocyte turnover per year in adulthood.¹⁹ However, the heart is unable to adequately compensate in the face of the overwhelming cardiomyocyte loss seen in AMI and HF. Stem cells, defined by their ability to self-renew and differentiate into at least 1 other cell type,²⁰ were investigated as a means to directly replace lost myocardium and stimulate endogenous repair. The current application of cell-based therapy to CVD is as an adjunct to evidence-based standard of care therapies to promote myocardial repair. Stem cells can be delivered through numerous methods to reach the target organ at different stages in the progression of heart disease (Figure 1).

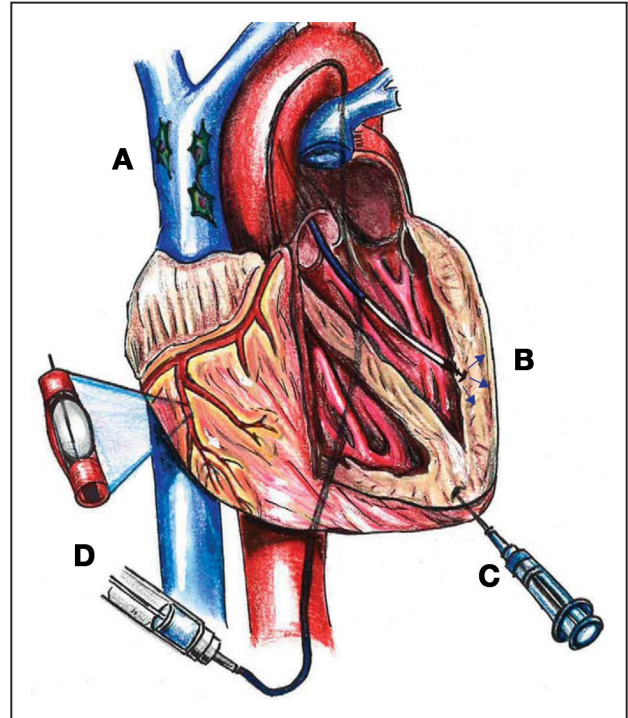


Figure 1. Modes of stem cell delivery. A, Intravenous delivery (peripheral veins not shown). B, Transendocardial injection (TESI) via catheter. C, Epicardial injection. D, Intracoronary infusion via catheter. Reproduced from Golpanian et al²¹ with permission. Copyright ©2016, the American Physiological Society.

Some are disillusioned by the small effect sizes and lack of definitive clinical benefit of many clinical trials using stem cells. Preclinical models indicated larger effects on surrogate end points, such as left ventricular ejection fraction (LVEF). According to meta-analyses, stem cell therapy improves LVEF by 7.5% in large animal models,²² and only 2% to 5%^{1,2,23,24} in clinical trials. However, these effect sizes come from meta-analyses of many trials of different cell types and patient populations, making such aggregate values difficult to interpret. Although some trials revealed smaller effect sizes than anticipated based on preclinical studies, there is still great promise for this therapy. What is clear from clinical experience is that most of the assessed cell types and delivery methods are safe and feasible, and many confer potential benefits for certain patient populations. To move the field forward, we must learn important lessons from previous trials. Future phase III trials require rigorous controls, blinding, patient selection, and larger sample sizes and should test the most effective, optimized cell types. This review critically examines the current state of clinical research on cell-based therapies in CRM, highlighting the controversies in the field, improvements in clinical trial design, and the application of exciting new cell products and technologies.

Modes of Action and Cell Types

Initial studies hypothesized that transplanted cells could directly differentiate into cardiomyocytes and therefore remuscularize the heart. However, while this can occur,^{25,26} it has proven to be a rare event.^{27,28} In fact, there is little evidence of substantial long-term engraftment of transplanted cells

despite durable improvements in cardiac function,²⁹ indicating that paracrine signaling is a crucial mode of action.^{21,30,31} Secretion of growth hormones, cytokines, metalloproteinases,²¹ exosomes, and mitochondrial transfer³² is all means by which transplanted cells can modulate the damaged tissue and stimulate endogenous repair mechanisms. Stem cells have also been shown to engraft and form gap junctions with cardiomyocytes in a process called heterocellular coupling that may impact electrophysiology; however, the importance of this phenomenon requires further investigation.^{32a} There are several processes that contribute to adverse cardiac remodeling that are potential targets for cell-based therapies, including imbalanced extracellular matrix turnover and fibrosis,³³ acute and chronic inflammation,³⁴ oxidative stress,³⁴ abnormal energy metabolism and oxygen consumption,³⁵ increased apoptosis,³⁶ endothelial dysfunction,³⁷ and reduced vasculogenesis³⁸ (Figure 2). The exact mechanisms of action for different cell types require further investigation but should not stall clinical trials as a detailed understanding of mechanism is not necessary for evaluating efficacy. However, we still depend heavily on preclinical studies to evaluate new stem cells as the field moves away from the use of first-generation cells, and continued preclinical investigation should occur simultaneously with carefully designed clinical trials.

So-called first-generation stem cells applied to clinical trials for CVD include those derived from adult sources (such as bone marrow and adipose tissue), delivered as unfractionated populations of cells or as purified, unmodified single cell types (Figure 3). Some of the earliest clinical trials used unfractionated bone marrow mononuclear cells (BMMNCs), a heterogeneous population comprising mesenchymal stem cells (MSCs), endothelial progenitor cells, and hematopoietic stem cells among others. BMMNCs can be obtained via bone marrow aspiration, do not require extensive expansion, and can be isolated with density gradient centrifugation.³⁹ Advances in the understanding of stem cell biology and isolation techniques led to the use of purified progenitor cell populations, such as MSCs⁴⁰ and CD34⁺ cells.⁴¹

MSCs are fibroblast-like cells and according to the tissue stem cell committee of the international society of cellular therapy are defined by their adherence to plastic under certain culture conditions, expression of cell markers CD73, CD90, CD105, absence of several others, and ability to be differentiated into adipocytes, chondrocytes, and osteoblasts in vitro.⁴² MSCs can be isolated from many tissues, including bone marrow, adipose tissue, umbilical cord blood, and even dental pulp.⁴³ They require expansion in vitro and therefore cannot be administered on the day of collection like BMMNCs, limiting their use as autografts for acute illnesses. MSCs possess potent immunosuppressive properties and secrete a wide variety of proangiogenic and antiapoptotic paracrine factors that stimulate cardiac repair.³⁹ In addition, they lack major histocompatibility type II markers, making them excellent for use as allografts.²¹ CD34 and CD133 are surface markers for both hematopoietic stem cells and endothelial progenitors, the latter of which are recruited to peripheral blood in response to injury, and when isolated from blood are often called circulating progenitor cells.³⁹ Circulating progenitor cells can give rise to blood cell types and endothelial cells and are proangiogenic.

The heterogeneous population of adipose-derived regenerative cells is obtained from liposuction samples, and the stromal fraction contains multipotent cells that can differentiate into several lineages.⁴⁴ These cells can be delivered the same day as collection and contain a high proportion of MSCs and CD34⁺ cells.⁴⁵

Second-generation cell therapies include progenitor cells isolated from cardiac tissue, allogeneic cell products, human pluripotent stem cells, and stem cells that have been lineage directed via chemical or genetic treatments to take on a specific cell phenotype (Figure 3). This group includes c-kit⁺ (CD117⁺) CSCs and cardiosphere-derived cells (CDCs). CSCs are isolated from adult hearts and display strong paracrine signaling and multilineage transdifferentiation in vitro.⁴⁶ CDCs, first described by Messina et al,⁴⁷ are a heterogeneous population of cells derived from myocardial tissue that form self-adherent clusters in vitro and include both CSCs and supporting cells among others.⁴⁸ Lineage-directed cells are those that have been treated to express a specific phenotype, such as the cardiopoietic cells used in the CHART-1 trial (The Congestive Heart Failure Cardiopoietic Regenerative Therapy).⁶ These bone marrow-derived MSCs (BM-MSCs) were treated with a cocktail of trophic factors to elicit a cardiac phenotype.⁶ Allogeneic cells are isolated from young, healthy donors and can be saved for later delivery and delivered as an off the shelf product, eliminating the need for long waiting periods required to expand autologous cell populations. There is also some evidence that cells from young, disease-free donors are superior to those from older patients with comorbidities, including ischemic heart disease⁴⁹ and atherosclerosis.⁵⁰ In addition, there is evidence that older patients respond similarly to allogeneic MSC therapy compared with younger ones⁵¹; however, there is also evidence that younger patients respond better to stem cell therapy, and this issue requires further investigation. Second-generation stem cells also include pluripotent cells, such as embryonic stem cells (ESCs) and induced pluripotent stem cells (iPSCs).⁵² These cells are unique in that they have a vast potential for differentiation, but to prevent formation of tumors, must be lineage directed before transplantation. Although iPSCs have not yet entered clinical use, the first human trial of lineage-directed ESCs for CVD was recently published.⁵³

End Point Evaluation in Clinical Trials

One of the major challenges in translating clinical trials of cell-based therapies to practice has been the relatively small sample sizes and reliance on surrogate end points in studies thus far. A surrogate end point is a functional or biochemical biomarker that substitutes for a clinical end point but is expected to predict clinical benefit based on evidence.⁵⁴ The use of surrogate end points can overcome financial and compliance constraints in clinical trials, allowing for reductions in sample size and follow-up periods. The rapid application of stem cells to clinical trials has outpaced our understanding of their modes of action, making it unclear exactly which surrogate end points to examine to make meaningful claims about efficacy. However, as the focus of cell therapy shifted toward chronic HF, the examination of a more diverse array of end points was adopted.

LVEF has long been considered to be an important functional surrogate end point for trials in patients with HF, largely because it predicts cardiovascular mortality.⁵⁵ In fact, in a study of 30 mortality trials of drug and device therapies, short-term

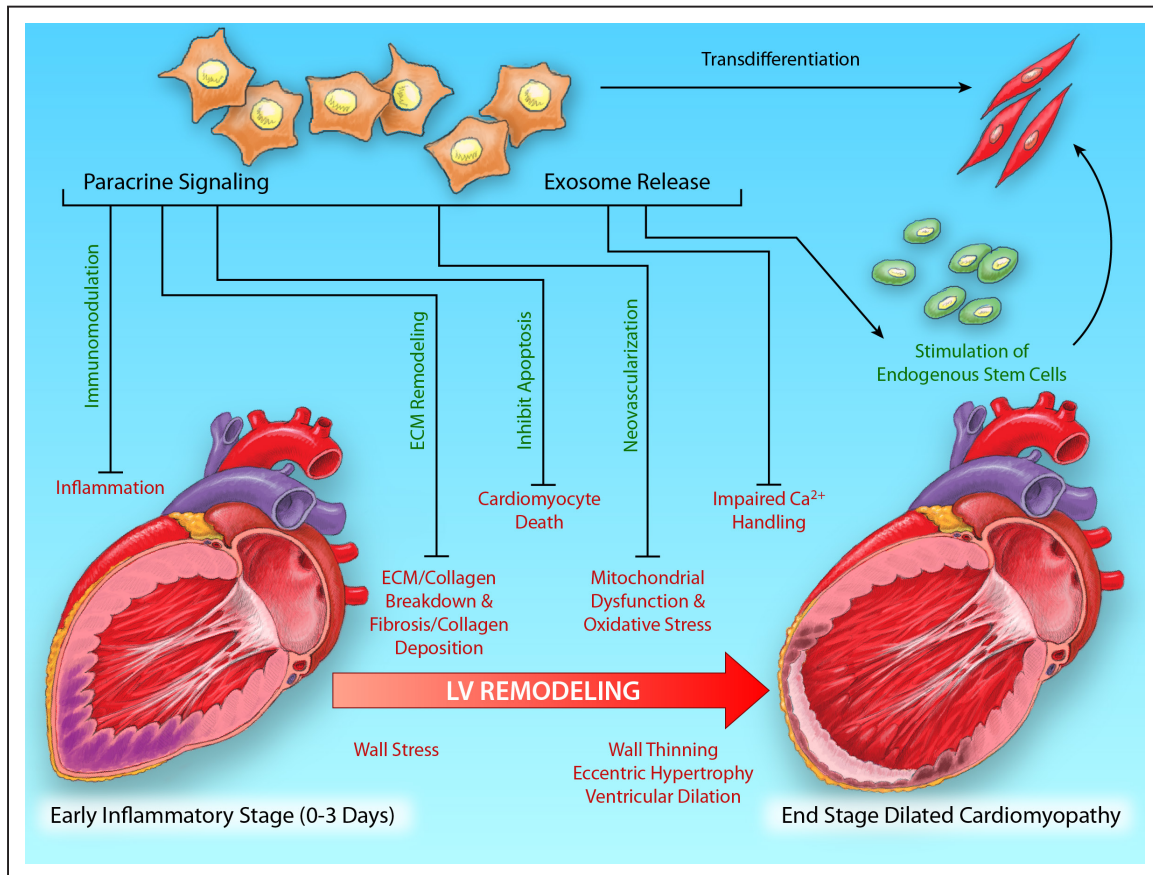


Figure 2. Left ventricular (LV) remodeling and potential targets for stem cell therapy. After an initial ischemic insult, the resulting necrosis, extracellular matrix (ECM) turnover, and collagen breakdown promote homing of inflammatory cells. Cardiomyocyte death forces the remaining cells to work harder to maintain cardiac output. The resulting cardiomyocyte hypertrophy, ECM remodeling, and fibrosis produce wall thinning, increased wall stress, ventricular dilation (remodeling), and impaired function. The overworked cardiomyocytes exhibit mitochondrial dysfunction, oxidative stress, and impaired Ca^{2+} handling. Stem cells release paracrine mediators and directly differentiate into cardiomyocytes, inhibit cardiomyocyte apoptosis and inflammation, promote angiogenesis, and stimulate endogenous cardiac stem cells. Together, these stem cell effects contribute to reversing/attenuating remodeling (Illustration credit: Ben Smith).

therapeutic effects detected in trials were associated with longer term trial-level effects on mortality.⁵⁶ Although some individual trials have reported profound improvements in LVEF, which may reflect the improvement in cardiomyocyte function as a result of cellular therapy, others have criticized the reliance on LVEF because it is load dependent, and as cardiac remodeling occurs, parallel decreases in left ventricular end-diastolic volume (LVEDV) and left ventricular end-systolic volume (LVESV) will result in an unchanged LVEF despite improved cardiac dynamics.⁵⁷ Similarly, measures of regional left ventricular (LV) function, including the wall motion score index, are also predictors of adverse events after myocardial infarction (MI).⁵⁸

Because LV remodeling is an independent predictor of cardiovascular mortality, it is an important surrogate marker for the clinical efficacy of cardiovascular therapies.⁵⁹ Remodeling is marked by progressive wall thinning, fibrosis, and LV dilation, which can be measured by LVESV and LVEDV, infarct size, and chamber dimensions.⁶⁰ Stem cells are thought to attenuate and even reverse the remodeling process by increasing perfusion to the stressed myocardium via neovascularization, stimulating new cardiomyocyte formation, and improving

existing cardiomyocyte survival and by reducing fibrosis through modulation of the extracellular matrix (Figure 2).²¹ As such, we would expect to see reductions in LV volumes, sphericity index, and infarct size as a result of stem cell reverse remodeling. Similarly, neoangiogenesis should improve myocardial perfusion as measured by single-photon emission computed tomography and cardiac magnetic resonance imaging (cMRI).

One of the most notable effects of some stem cells is a significant reduction in infarct size, which has high validity as a surrogate end point. Scar size is a predictor of cardiac events, and because it is independent of loading, directly relates to LV remodeling.⁶¹ In clinical trials, MSCs reduce scar size at 1 year by $\approx 38\%$ compared with scar size at baseline.⁶² cMRI cannot distinguish between myocardial hyperplasia and new cell formation, or whether scar size is reduced simply by thinning, therefore the reported increases in myocardial mass have been critiqued as just hypertrophy of existing cardiomyocytes rather than true regeneration.⁶³ However, multiple large animal studies have demonstrated the formation of new cardiomyocytes at the border zones of infarcts and have failed to show evidence of myocyte hypertrophy, supporting

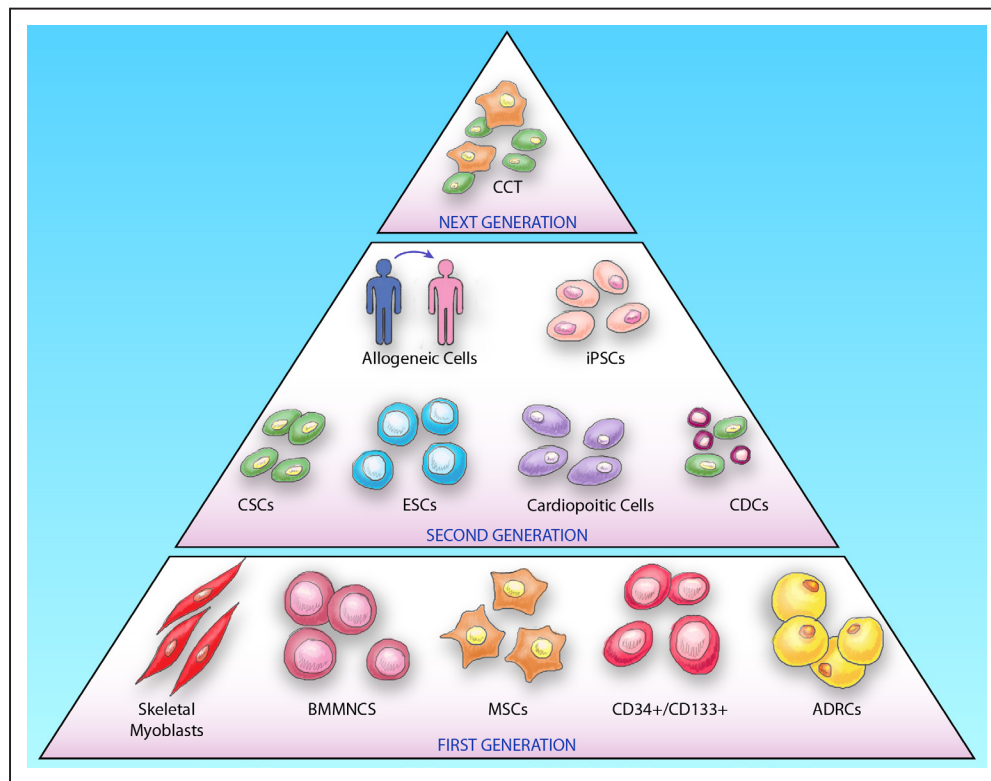


Figure 3. Advances in cardioregenerative medicine. First-generation stem cell therapy used skeletal myoblasts, heterogeneous populations of cells, such as bone marrow mononuclear cells (BMMNCs) and adipose-derived regenerative cells (ADRCs), and more purified mesenchymal stem cells (MSCs) and hematopoietic stem cells (CD134⁺/CD133⁺) isolated from either bone marrow or blood. Second-generation stem cells include purified cardiac cell populations such as ckit⁺ cardiac stem cells (CSCs) and cardiosphere-derived cells (CDCs), pluripotent cells (embryonic stem cells [ESCs] and induced pluripotent stem cell [iPSCs]), allogeneic cells. Next-generation stem cells include modified, cells, and lineage-directed stem cells, such as cardiopoietic stem cells derived from MSCs. The next generation of cell therapies includes cell combination therapy (CCT) of specific cell types (Illustration credit: Ben Smith).

a mode of action in which stem cells promote endogenous myogenesis.^{25,64,65}

Although mortality and its surrogate predictors including LVEF and LV remodeling are the most commonly assessed end points in clinical trials of HF, patient functional status and quality of life (QoL) are also important markers as they are impaired in HF reflecting the symptomatic burden of the HF syndrome.⁶⁶ Functional capacity includes measures of exercise ability and cardiorespiratory fitness and can be measured objectively by parameters, such as the 6-minute walk test and maximal oxygen consumption, both of which predict mortality in patients with HF.⁶⁷ Several tools assess QoL among patients with HF, including the Minnesota Living with Heart Failure Questionnaire (MLHFQ) and the Kansas City Cardiomyopathy Questionnaire, and although there is debate as to whether these are reliable markers of health status, they are important aspects of a patient's experience of symptoms and may improve along with reversal of HF as they do in patients on angiotensin-converting enzyme inhibitors and β -blockers.⁶⁸ Markers of symptoms and functionality, such as New York Heart Association (NYHA) classification and the Canadian Cardiovascular Society score for angina, are extremely important in assessing the impact of cell therapy on symptoms, especially in the setting of refractory angina.

Serum biomarkers are increasingly used in cardiovascular research and are being incorporated into cell therapy trials.

In HF, the natriuretic peptides, including B-type natriuretic peptide and N-terminal pro-B-type natriuretic peptide, are important diagnostic and prognostic tools and are predictors of mortality. Furthermore, these markers decline in response to standard HF therapy, and B-type natriuretic peptide- and N-terminal pro-B-type natriuretic peptide-guided therapy reduces hospitalizations and mortality, confirming their validity as surrogate markers.⁶⁹ A more diverse array of biomarkers may allow for precise delineation of the mode of action of stem cells in cardiac disease. For example, markers of extracellular matrix remodeling, such as collagen degradation products, could be used to assess remodeling. Inflammatory markers are elevated in HF and may reflect continued immune dysfunction and oxidative damage in the myocardium,⁷⁰ therefore measurements of TNF (tumor necrosis factor)- α and C-reactive protein may be important surrogate end points in trials of CRM. In the TRIDENT trial (Transendocardial Stem Cell Injection Delivery Effects on Neomyogenesis), 30 patients were randomized to receive transendocardial stem cell injection (TESI) of allogeneic MSCs at either a dose of 20 or 100 million cells. Although the results indicate that only those who received 100 million cells experienced improvements in LVEF, both groups experienced a significant reduction in TNF- α , both in the circulation and intracellular in B-cell, indicating the immunomodulatory effects of MSCs, which may play an important role in their improving cardiac function.⁷¹

A diverse array of surrogate markers is needed in early phase clinical trials to garner insight into the efficacy of specific cell types and inform phase III trial design. Early trials evaluated fewer parameters; however, more modern trials incorporate assessment of cardiac function and structure, QoL, and functional capacity. Finally, trials are evolving away from reliance on a single primary end point toward composite end points or consideration of multiple individual end points. There is, however, no substitute for long-term evaluation of hard clinical end points, such as mortality, hospitalizations, and cardiac events, which requires adequately powered, longitudinal, placebo-controlled studies.

Acute Myocardial Infarction

The application of cell-based therapy in clinical medicine largely began with trials investigating AMI. The goal was to use intracoronary infusion of stem cells as an adjunct to percutaneous coronary intervention to limit cardiomyocyte necrosis and prevent progression to HF, a strategy also known as cardioprotection.⁷² The most commonly used product in this setting is BMMNCs, harvested from patients 1 to 12 days after percutaneous coronary intervention and delivered the same day. BMMNCs have the advantage of not requiring culture expansion and can be harvested, isolated, and re-infused quickly. The early TOPCARE-AMI (Transplantation of Progenitor Cells and Regeneration Enhancement in Acute Myocardial Infarction)^{73,74} and BOOST⁷⁵ (Bone Marrow Transfer to Enhance ST-Elevation Infarct Regeneration) trials both evaluated BMMNCs delivered an average of 5 days after AMI. Both trials reported improvements in LVEF in the BMMNC-treated groups during a short follow-up period (4–6 months), as well as improvements related directly to cardiac remodeling, including infarct size reduction, spurring enthusiasm for BMMNC therapy. It must be noted, however, that TOPCARE-AMI did not include a control group, and although the BOOST trial did compare cell-treated patients to control patients receiving the standard of care, this study was by nature open label because the control patients did not undergo any additional procedures. The Leuven-AMI⁷⁶ and FINCELL⁷⁷ (Finish Stem Cell study) trials, which were double blinded, also reported improvements in LVEF in response to BMMNC infusion.

The REPAIR-AMI study (The Reinfusion of Enriched Progenitor Cells and Infarct Remodeling in Acute Myocardial Infarction),⁷⁸ a phase III, double-blinded, placebo-controlled clinical trial, is the largest clinical trial for AMI to date and was designed to assess the efficacy of BMMNCs. This study randomized patients (n=204) to receive either cells or placebo 3 to 7 days after AMI. At 4 months, LVEF was significantly improved by an average 5.5% in the BMMNC group compared with only 3.0% in the placebo group.⁷⁸ At 1 year, the composite of death, MI, and need for repeat revascularization were lower in the BMMNC group,⁷⁹ and this effect on mortality, but not the improvement in LVEF, was maintained at 5 years.⁸⁰

Despite these initial encouraging findings, other trials have produced contradictory results, tempering the excitement regarding BMMNCs (Table 1). The multicenter, double-blinded, placebo-controlled TIME (Use of Adult Autologous Stem

Cells in Treating People Who Have Had a Heart Attack)⁸¹ and LateTIME (Use of Adult Autologous Stem Cells in Treating People 2 to 3 Weeks After Having a Heart Attack)⁸² trials performed by researchers of the cardiovascular cell therapy research network (CCTRN) in the United States found no improvements in LVEF, LV volumes, or wall motion as measured by cMRI, between the BMMNC and placebo groups. Two-year follow-up of the 120 patients from the TIME study showed no improvements over placebo in LV function as measured by cMRI.⁸⁶ The SWISS AMI trial (Swiss Multicenter Intracoronary Stem Cells Study in Acute Myocardial Infarction),⁸³ which examined BMMNC administration at 2 intervals post-AMI (5–7 days and 3–4 weeks) in 192 patients, found no difference in cardiac function, cardiac volumes, or scar size compared with controls. The recent BOOST-2 trial⁸⁴ examined the effect of BMMNCs delivery 1 to 2 days after percutaneous coronary intervention in 153 patients and was unable to repeat the positive findings found in the original BOOST trial. The investigators compared high and low doses of placebo, BMMNCs, and γ -irradiated BMMNCs. The last group was added because γ -irradiation inhibits mitosis and prevents stem cell differentiation, allowing the investigators to test if mitosis is necessary for efficacy. There was no difference in LVEF, LV volumes, or scar mass at 6 months (as measured by cMRI) between any of the groups.⁸⁴ The most recently published randomized, double-blinded placebo-controlled trial investigating BMMNCs for AMI, the MiHeart/AMI trial⁸⁷ (n=121) also found no improvements in markers of LV remodeling or LVEF. Although REPAIR-AMI demonstrated improvements on hard clinical end points, meta-analysis of individual patient data (n=1252)⁸⁸ and aggregate data² from randomized control trials in AMI found no benefit of intracoronary cell therapy on clinical events or LVEF.

Given that BMMNCs are an unselected population, the selected subpopulation of CD34+ endothelial progenitors have also been examined as therapeutics for AMI because of their increased angiogenic capacity.⁸⁹ The PreSERVE-AMI study (The NBS10 Versus Placebo Post ST Segment Elevation Myocardial Infarction),⁸⁵ the largest trial of stem cells for AMI in the United States, examined CD34+ cells isolated from bone marrow compared with placebo. At 1 year, cell therapy did not improve LVEF or resting myocardial perfusion; however, when adjustments were made for the duration of ischemia, there was a dose-dependent improvement in LVEF, infarct size, and survival, indicating that this purified population of cells may exert beneficial effects.⁸⁵

Early trials in AMI are marked by a large degree of heterogeneity, highlighting the importance of rigor and standardized techniques in clinical trials. Cell handling and isolation varied greatly between trials, which impacts the total cell number, colony-forming units, and migratory capacity.⁹⁰ These findings further highlight the heterogeneity of BMMNCs and the impact of different isolation and storage media on their biological activity and support the need for assays to determine the function of cells before their use in clinical trials, a concept that still has yet to reach practice. Future trials must adhere to strict cell handling techniques that are clear and reproducible.

Imaging techniques also varied between studies. In the early, positive trials, LVEF and cardiac function were assessed

Table 1. Clinical Trials in Acute Myocardial Infarction

Trial	Follow-Up	n	Timing of Cell Delivery After MI	Cell Type and Dose	Evaluation Method	LVEF	Volumes	Scar Size
TOPCARE-AMI Schächinger et al ⁷⁴	12 mo	59	5 d	CPCs; BMMNCs 7.3×10 ⁶	LVG	↑	↓ ESV	↓
BOOST Wollert et al ⁷⁵	6 mo	60	5 d	2.4×10 ⁹ BMMNCs vs control	cMRI	↑	NS	NA
REPAIR-AMI Schächinger et al ⁷⁸	4 mo	204	3–5 d	198×10 ⁶ BMMNCs vs placebo	LVG	↑	NS	NA
TIME Traverse et al ⁸¹	6 mo	120	3 vs 7 d	150×10 ⁶ BMMNCs vs placebo	cMRI	NS	NS	NS
LateTime Traverse et al ⁸²	6 mo	87	2–3 wk	150×10 ⁶ BMMNCs vs placebo	cMRI	NS	NS	NS
SWISS AMI Sürder et al ⁸³	12 mo	192	5–7 d vs 3–4 wk	153×10 ⁶ BMMNCs vs control	cMRI	NS	NS	NS
BOOST-2 Wollert et al ⁸⁴	6 mo	188	7 d	20.6×10 ⁸ vs 7.0×10 ⁸ BMMNCs vs placebo	cMRI	NS	NS	NS
Pre-SERVE-AMI Quyyumi et al ⁸⁵	12 mo	161	4–11 d	14.9×10 ⁶ CD34 ⁺ vs placebo	cMRI	NS	NS	NS

↑ indicates increase; ↓, decrease; AMI, acute myocardial infarction; BMMNCs, bone marrow mononuclear cells; CD34⁺, hematopoietic stem cells; cMRI, cardiac magnetic resonance imaging; CPC, circulating progenitor cells; LVEF, left ventricular ejection fraction; LVG, left ventriculography; n, number of patients; NA, not assessed; and NS, not significant.

using either left ventriculography or echocardiography, both of which are fraught with issues of interobserver variability and precision.⁹¹ cMRI is widely regarded as the gold standard for cardiac imaging because of its higher precision and ability to measure scar tissue with the use of late gadolinium enhancement.⁹¹ In a meta-analysis of AMI trials, LVEF was significantly improved by BMMNC therapy in studies using echocardiography or left ventriculography but not those using single-photon emission computed tomography or cMRI.⁹² Another meta-analysis using only cMRI results showed no significant benefit of BMMNC administration after AMI on LVEF, LV volumes, or infarct size.⁵⁷ These findings are consistent with more recent trials using cMRI that failed to show a benefit of BMMNCs.^{81–84,93}

Over 15 years of clinical experience has provided important lessons for the field. Consistency in measurement of end points has improved, with most investigations adopting cMRI for anatomic and functional assessment. Similarly, standardization in techniques for cell isolation and handling has been championed by consortiums like CCTRN. With these improvements, the most recent trials using BMMNCs for AMI have failed to show any benefit. It is possible that the inflammatory, hypoxic postinfarct environment may be lethal to most transplanted cells, limiting their survival and reparative efficacy. The BAMI trial (Effect of Intracoronary Reinfusion of Bone Marrow-Derived Mononuclear Cells [BM-MNC] on All Cause Mortality in Acute Myocardial Infarction; NCT01569178) was designed to definitively evaluate the efficacy of BMMNCs in the setting of AMI. This ongoing multicenter, randomized, double-blinded placebo-controlled phase III trial aimed at enrolling 3000 patients in an attempt to discern impacts on hard clinical end points, such as mortality; however, recently the group reported that the trial is now expected to recruit only 350 patients.⁵ Although 350 patients

would make this study the largest stem cell study for AMI to date, the low recruitment is disappointing as it may not answer definitively if we should cease using BMMNCs for AMI.

BMMNCs have not shown much efficacy in recent trials, prompting endeavors to find more efficacious cell types, and MSCs seem to be good candidates. In fact, trials assessing other cell types, such as allogeneic MSCs for AMI, have produced promising results. Hare et al⁹⁴ reported that intravenous administration of allogeneic MSCs was safe and reduced episodes of ventricular tachycardia and improved LVEF, especially in patients with anterior MI. These initial results will be tested in a 220-patient phase II study (NCT00877903). Gao et al⁹⁵ randomized 116 patients to either intracoronary placebo or umbilical cord–derived MSC (UC-MSC) treatment 5 to 7 days after AMI and noted reductions in scar size and a mean increase in absolute LVEF of 7.8% (5% greater than in placebo). The ongoing CAREMI trial (Safety and Efficacy of Intracoronary Infusion of Allogeneic Human Cardiac Stem Cells in Patients with AMI) is a phase I/II trial assessing allogeneic CSCs in 55 patients with AMI versus control (NCT02439398),⁹⁶ and the AMICI trial (Safety Study of Allogeneic Mesenchymal Precursor Cell Infusion in Myocardial Infarction) is a phase II trial examining intracoronary delivery of mesenchymal precursor cells (NCT01781390).

Refractory Angina

Refractory angina is a growing problem among survivors of ischemic cardiac events with up to 1.8 million patients in the United States.³ These clinically challenging patients are usually ineligible for or do not benefit from further revascularization because of microvascular disease and there are few effective treatments available for their symptoms.⁹⁷ Stem cells are an attractive therapeutic option for these patients because they stimulate neoangiogenesis to improve perfusion.⁹⁸

BMMNCs have been delivered to this patient population, either through coronary infusion or TESI. Similarly, purified populations of CD34+ and CD133+ cells have been tested given their improved capacity for angiogenesis and improved perfusion.⁸⁹ Losordo et al⁹⁸ investigated endomyocardial injection of a low or high dose of CD34+ circulating progenitor cells versus placebo in 167 patients with refractory angina. Angina frequency was significantly lower in the cell-treated group compared with the placebo group after 12 months (6.3 versus 11 events/wk, respectively). Furthermore, there was a significant improvement in exercise tolerance in those patients receiving the lower cell dose.

The RENEW trial (Efficacy and Safety of Targeted Intramyocardial Delivery of Auto CD34+ Stem Cells for Improving Exercise Capacity in Subjects With Refractory Angina)⁹⁷ was initially planned as a phase III trial to assess the efficacy of autologous CD34+ cells isolated from peripheral blood. The trial fell short of its projected 444 patient enrollment because of termination for financial considerations, enrolling only 112 patients. Despite the small size, the trial did demonstrate improvements in total exercise time over placebo at 3 months and angina frequency at 6 months in patients receiving endomyocardial cell injections compared with no intervention. There was no significant improvement in exercise time in the cell-treated patients at 6 and 12 months. However, a patient-level pooled analysis

of 304 patients from 3 double-blinded, placebo-controlled randomized control trials of TESI of CD34+ cells found that therapy resulted in clinically meaningful and lasting (up to 12 months) improvements in total exercise time and angina frequency. Furthermore, there was reduced mortality in the treated population at 24 months.⁹⁹ Another meta-analysis of 6 trials using stem cells for refractory angina supports CD34+ cell therapy as it reduces the number of angina episodes and improves exercise tolerance and myocardial perfusion. Taken together, these studies indicate that CD34+ cell therapy may provide a clinically meaningful solution in this patient population, even without definitive evidence of reduced mortality from a single trial.³

Ischemic Cardiomyopathy

Ischemic heart disease is the leading cause of death in the United States and worldwide.¹² After MI, the extent of injury correlates with the degree of cardiac remodeling, which in turn is the substrate for HF and sudden cardiac death.¹⁰⁰ Because CRM aims to repair tissue injury and restore the function of damaged myocardium, it represents a potentially curative rather than palliative treatment for chronic ischemic cardiomyopathy (ICM). ICM, being a chronic disorder, allows time for the harvesting, purification, and in vitro expansion of autologous cells. Therefore, trials in ICM (Table 2) are notable for the use of a more diverse array of

Table 2. Clinical Trials in Ischemic Cardiomyopathy

Trial	Follow-Up	n	Cell Type, Dose, and Treatment Groups	Delivery Method	End Point Evaluation	LVEF	LV Volumes	Scar Size	NYHA Class	Functional Capacity	QoL
Perin et al ^{101,102}	4 mo	21	25.5×10 ⁶ BMMNCs vs control	TESI	Echo	↑	↓ESV	NA	NA	↑	NA
TOPCARE CHD Assmus et al ¹⁰³	3 mo	75	22×10 ⁶ CPCs vs 205×10 ⁶ BMMNCs vs control	IC	LVG	BMMNC: ↑EF	NS	NS	BMMNC: ↑	NA	NA
FOCUS CCTRN Perin et al ⁷	6 mo	92	100×10 ⁶ BMMNCs vs placebo	TESI	Echo SPECT	NS	NS	NS	NS	NS	NA
MSC-HF Mathiasen et al ¹⁰⁴	6 mo	55	77×10 ⁶ BM-MSC vs placebo	TESI	cMRI or CT	↑	↓ESV	↓	NS	NS	NS
SCIPPIO Bolli et al ¹⁰⁵	4 mo	23	1×10 ⁶ CSCs vs control	IC	Echo/cMRI	↑	NA	↓	↓	NA	↓
POSEIDON Hare et al ¹⁰⁶	13 mo	30	20, 100, or 200×10 ⁶ BM-MSCs (allo vs auto)	TESI	cMRI	NS	Allo: ↓EDV	↓	NS	Auto: ↑	Auto: ↑
CADUCEUS Makkar et al ¹⁰⁷	6 mo	31	12.5–25×10 ⁶ CDCs vs control	IC	cMRI	NS	NS	↓	NS	NS	NS
CHART-1 Bartunek et al ⁶	39 wk	351	9.7×10 ⁸ –1.2×10 ⁹ Cardiopoietic cells vs sham	TESI	Echo	NS	NS	NA	NA	NS	NS
TAC-HFT Heldman et al ⁴⁰	12 mo	65	100×10 ⁶ BM-MSCs vs 100×10 ⁶ BMMNCs vs placebo	TESI	cMRI/CT	NS	NS	MSC: ↓	NS	MSC: ↑	↑
RIMECARD Bartolucci et al ¹⁰⁸	12 mo	30	1×10 ⁶ /kg UC-MSC vs placebo	IV	Echo/cMRI	↑	↑EDV	NA	↑	↑	↑
ATHENA Trials Henry et al ⁴⁵	12 mo	31	80×10 ⁶ vs 40×10 ⁶ ARDCs vs placebo	TESI	Echo	NS	NS	NA	↑	↑	↑

↑ indicates increase; ↓, decrease; allo, allogeneic; ARDCs, adipose-derived regenerative cells; auto, autologous; BMMNCs, bone marrow mononuclear cells; BM-MSCs, bone marrow–derived mesenchymal stem cells; CCTRN, cardiovascular cell therapy research network; CDCs, cardiosphere-derived cells; cMRI, cardiac magnetic resonance imaging; CPCs, circulating progenitor cells; CT, computed tomography; echo, echocardiography; EDV, end-diastolic volume; EF, ejection fraction; ESV, end-systolic volume; IC, intracoronary; IV, intravenous; LV, left ventricle; LVEF, left ventricular ejection fraction; n, number of patients; NA, not assessed; NS, not significant; NYHA, New York Heart Association; QoL, quality of life; SPECT, single-photon emission tomography; TESI, transendocardial stem cell injection; and UC-MSCs, umbilical cord–derived mesenchymal stem cells.

cell types, including purified, culture-expanded subpopulations of BMMNCs (such as MSCs) and the introduction of second-generation cells (Figure 3). In addition, lessons learned from large clinical trials of pharmaceuticals for HF encouraged broadening of the surrogate end points evaluated to include cardiac function and anatomy, QoL, and functional capacity.

The clinical application of cell therapy for CVD began in the setting of ICM in 2001, with transplantation of skeletal myoblasts into damaged myocardium at the time of coronary artery bypass grafting.⁴ Despite the initial enthusiasm for these cells and their purported ability to engraft and function within the ailing myocardium, they have since been abandoned because of the increased ventricular tachyarrhythmias and lack of efficacy.¹⁰⁹ These cells lack gap junction expression after differentiation, thereby preventing electrical coupling to resident cardiomyocytes and increasing the propensity for arrhythmias.¹¹⁰

Bone Marrow Mononuclear Cells

Perin et al¹⁰¹ were among the first to assess the safety and efficacy of BMMNCs in patients with HF. Cells were delivered via percutaneous TESI using the NOGA Myostar catheter, which allows for electromechanical mapping to accurately delineate the area of infarct and guide injection into the border zones. BMMNC injection improved LVEF by 9% and reduced LVESV and scar size.¹⁰¹ Furthermore, exercise capacity was significantly improved at 6 and 12 months. In the largest trial of BMMNC therapy in patients with ICM, Pokushalov et al assessed TESI of BMMNCs versus usual treatment in 109 patients with severe chronic HF (LVEF $\leq 35\%$) secondary to ischemic HF secondary to MI >12 months before treatment. Compared with controls, the BMMNC group (n=55) experienced significantly greater improvements in Canadian Cardiovascular Society angina class, NYHA class, and absolute LVEF (by 4.5%) 6 months after treatment. Importantly, there were significantly fewer deaths in the cell-treated group a year after randomization, indicating that BMMNC therapy may improve survival in this population of HF patients with no further options for revascularization.

By contrast, only minimal improvements in LVEF were seen in the phase II, double-blinded, randomized FOCUS CCTR trial (Effectiveness of Stem Cell Treatment for Adults With Ischemic Cardiomyopathy),⁷ which compared placebo and BMMNC therapy in 92 patients with ICM. At 6 months, there were no differences between cell- and placebo-treated patients in LVESV index, maximal O₂ consumption, or myocardial perfusion defect. Absolute LVEF improved by 1.2% in the BMMNC-treated patients and decreased by 1.3% in the placebo group. Further analysis revealed that improvements in LVEF were correlated with the proportion of CD34+ and CD133+ progenitor cells in the BMMNC samples, suggesting that this specific fraction of BMMNCs is associated with improved therapeutic efficacy.

Collectively, clinical trials using BMMNCs for ICM reveal positive effects of these cells on cardiac function and remodeling. In their meta-analysis, Xiao et al¹¹¹ report a significant improvement in absolute LVEF by a mean of 4.33% in trials using BMMNCs for ICM, along with reductions in

LVESV and LVEDV. These cells are relatively easy to isolate and administer in an autologous fashion, and they continue to be studied; however, the variation in results may be secondary to reduced efficacy in cells derived from older patients, as well as the specific subtypes of cells in the heterogeneous mix. As a result, most clinical trials currently recruiting ICM patients are investigating more purified cell types, allogeneic transplants, and combinations of cells, building on exciting findings from recent phase I/II trials.

Mesenchymal Stem Cells

Based on the growing preclinical evidence that MSCs are a key active constituent of the bone marrow cells that participate in cardiac repair, several clinical trials were conducted using MSCs for ICM. BM-MSCs are excellent candidates for CRM because they exhibit a 4-fold constellation of phenotypic activity that include immunomodulation, antifibrotic effects, stimulation of neoangiogenesis, and activation of endogenous cardiac repair pathways.²¹ These effects are attributable to the secretion of numerous cytokines and trophic factors,¹⁰⁰ release of exosomes, transfer of mitochondria,³² and formation of gap junctions with host cells.^{25,112} The antifibrotic effects result from secretion of matrix metalloproteinases and other molecules that modulate the extracellular matrix, thereby reducing the infarct size and diffuse fibrosis seen in ischemic HF.¹¹³ One of the earliest trials assessing MSCs for CVD was the phase II randomized, placebo-controlled TAC-HFT trial (Transendocardial Autologous Cells [hMSC or hBMC] in Ischemic Heart Failure),⁴⁰ which compared TESI of autologous BMMNCs, MSCs, and placebo. Although there was no significant improvement in LV volumes or LVEF in response to injection of either cell type, QoL as measured by the MLHFQ score, improved significantly from baseline in both treatment groups. The 6-minute walk distance (6MWD) improved only in the MSC group. Infarct size was reduced by 19% in the MSC group but remained unchanged in the BMMNC and placebo groups (Figure 4). These findings suggest that MSCs may have improved antifibrotic efficacy compared with BMMNC, and this impact on remodeling (via scar size reduction) also has an effect on exercise capacity. Furthermore, the TAC-HFT results demonstrate that even in the absence of statistically significant changes in LVEF and volumes, there can be improvements in both QoL and functional capacity. In the randomized, double-blinded TRIDENT trial,⁶² dosing of allogeneic MSCs was examined, and the higher 100 million cell dose significantly improved absolute LVEF by 3.7% compared with no change in 20 million group after 12 months. However, these findings require substantiation in larger clinical trials that also assess hard clinical end points.

MSCs are immunoprivileged in that they lack major histocompatibility complex class II molecules and secrete anti-inflammatory factors, making them particularly suitable for use as allografts. The POSEIDON trial (Percutaneous Stem Cell Injection Delivery Effects on Neomyogenesis)¹⁰⁶ was the first to compare autologous and allogeneic MSCs in patients with ICM. This randomized, double-blinded phase I/II trial examined the efficacy and safety of TESI of 3 different doses of these MSCs and evaluated a diverse array of end points in

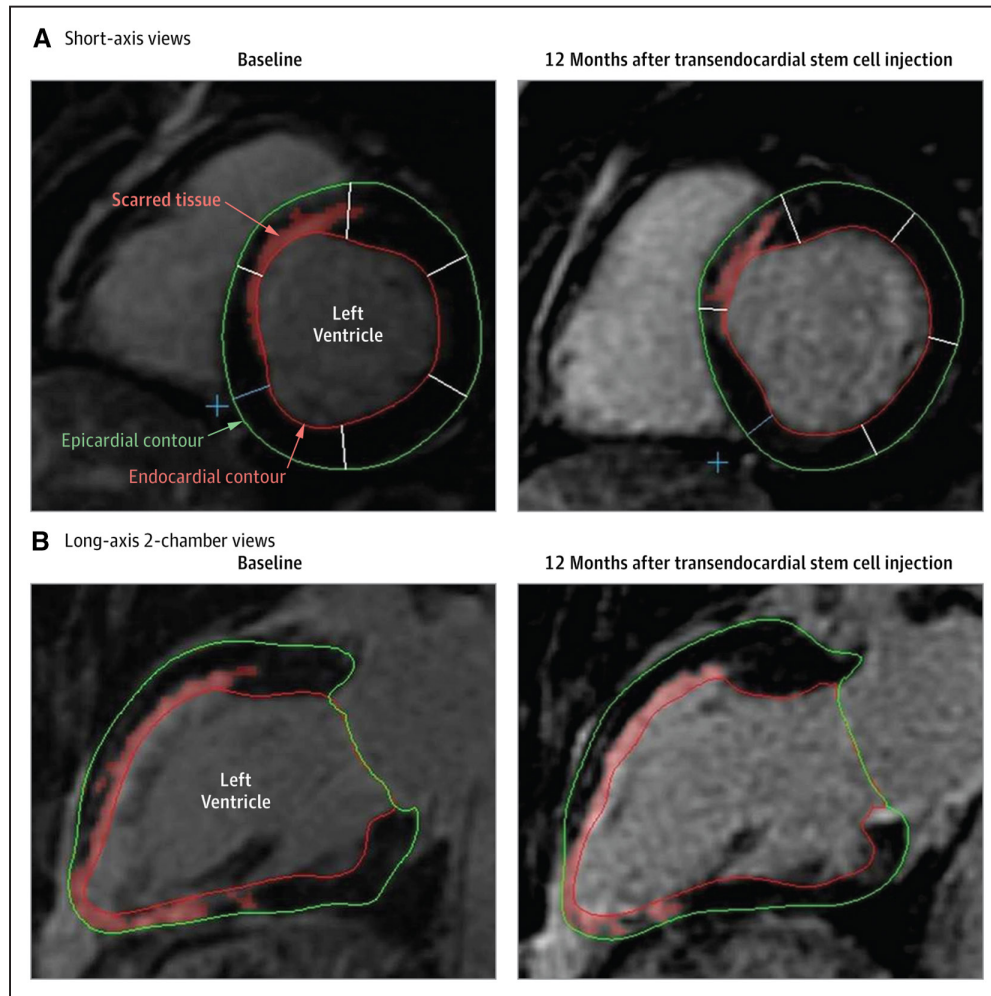


Figure 4. Scar size reduction in a TAC-HFT trial patient. Cardiac magnetic resonance images from a representative patient before and 12 mo after mesenchymal stem cell (MSC) Injection. **A**, Short-axis views of the basal area of a patient's heart, with delayed tissue enhancement delineated at the septal wall. Delayed tissue enhancement corresponds to scarred tissue and is brighter than the nonscarred tissue (automatically detected and delineated with red using the full width at half maximum technique). The red, green, and white lines demarcating the endocardial, epicardial contours, and borders of the segments, respectively, were drawn manually. Twelve months after injection of mesenchymal stem cells, scar mass was reduced significantly. **B**, Long-axis 2-chamber views of the same heart with delayed tissue enhancement delineated at the anterior and inferior wall, as well as the entire apex. At baseline and at 12 mo after injection of MSCs, the delayed tissue enhancement receded in the midinferior and basal anterior walls. Reproduced from Heldman et al⁴⁰ with permission. Copyright ©2014, American Medical Association.

order to further guide the selection of appropriate efficacy end points in phase III trials. From baseline to 1 year, both allogeneic and autologous MSCs reduced scar size by 33% as well as sphericity index, suggesting a marked impact on LV remodeling. Interestingly, while the LVESV and LVEDV decreased in both groups, a significant reduction was only seen in the LVEDV of the allogeneic group, which may be due increased potency of MSCs derived from young, healthy donors as compared with those from older patients with ICM. These studies again illustrates that even without improvements in ejection fraction, there were important reverse remodeling effects.

The MSC-HF trial assessed TESI of autologous MSCs in the setting of severe ICM in 55 patients with LVEF <45% and NYHA class II or III. At 6 months, there was a significant increase in LVEF by 6.2% in the MSC group compared with placebo, which was accompanied by reduced LVESV but not LVEDV. This marked impact on LVEF may be due to the higher baseline severity of ICM and differences in

cell selection and dose compared with the TAC-HFT and POSEIDON trials. Taken together, the results from these 3 studies indicate that BM-MSC are safe and may have an impact on LVEF, cardiac remodeling, functional capacity, and QoL. The ongoing, multicenter, phase III DREAM HF-1 trial (Efficacy and Safety of Allogeneic Mesenchymal Precursor Cells [Rexlemestrol-L] for the Treatment of Heart Failure; NCT02032004) is examining TESI of allogeneic mesenchymal precursor cells, a population of multipotent stromal cells isolated from bone marrow, not on their ability to adhere to plastic (as with MSCs), but via immunoselection based on certain markers, including Stro-1 and Stro-3. This population contains a higher number of colony-forming unit-fibroblasts and displays similar trilineage differentiation in defined culture conditions as MSCs.¹¹⁴ Mesenchymal precursor cells display higher doubling capacity and secretion of paracrine factors in vitro compared with density gradient-selected MSCs¹¹⁵ and are therefore

hypothesized to be more potent in treating cardiac disease. The study aims to enroll 600 participants with HF of either ischemic or nonischemic origin, and will compare TESI of mesenchymal precursor cells to sham controls on a number of surrogate and hard clinical end points.

The majority of clinical trials have focused on MSCs from bone marrow; however, stem cells from adult tissues may have limited paracrine and differentiation potential, which may be overcome by using UC-MSCs.¹¹⁶ Allogeneic UC-MSCs were assessed in the RIMECARD trial (Randomized Clinical Trial of Intravenous Infusion of Umbilical Cord Mesenchymal Stem Cells on Cardiomyopathy),¹⁰⁸ a phase I/II randomized, double-blinded, placebo-controlled trial investigating intravenous infusion of allogeneic UC-MSCs in 30 patients with HF (70% of nonischemic origin) with 12-month follow-up. The UC-MSC-treated group exhibited a significant 7% improvement in LVEF as measured by echocardiography. NYHA class and MLHFQ scores improved only in the UC-MSC group. One of the most interesting things about this trial is the rigorous characterization of the utilized cells including their differentiation potential which was tested in comparison to BM-MSCs. The UC-MSCs paracrine profile was evaluated in vitro and found to have higher *TGF- β 3* gene expression and hepatocyte growth factor expression compared with BM-MSCs. UC-MSCs were

also cocultured with T cells and found to diminish T cell proliferation in response to stimulation, supporting the anti-inflammatory effect which has been noted in preclinical studies.¹⁰⁸

Adipose-Derived Regenerative Cells

Adipose-derived regenerative cells have been tested in clinical trials as well, with some mild benefits. In the PRECISE trial (Randomized Clinical Trial of Adipose-Derived Stem Cells in Treatment of Non Revascularizable Ischemic Myocardium), 21 patients with ICM were treated with TESI of autologous adipose-derived regenerative cells and compared with 6 controls. Adipose-derived regenerative cells demonstrated safety and feasibility.⁴⁴ The larger trials⁴⁵ assessed injection of 2 doses (40 and 80 million cells) of autologous adipose-derived regenerative cells versus placebo. At 12 months, there were no differences in NYHA or Canadian Cardiovascular Society classes, LVEF, or LV volumes; however, there was an improvement in MLHFQ in the treatment group.⁴⁵ Adipose-derived stromal/stem cells are currently being evaluated in phase II clinical trials (NCT02673164).

Cardiac Stem Cells

Like MSCs, CSCs secrete cytokines and growth factors that stimulate endogenous stem cells and repair mechanisms.³⁹ Despite low engraftment and limited maturation into

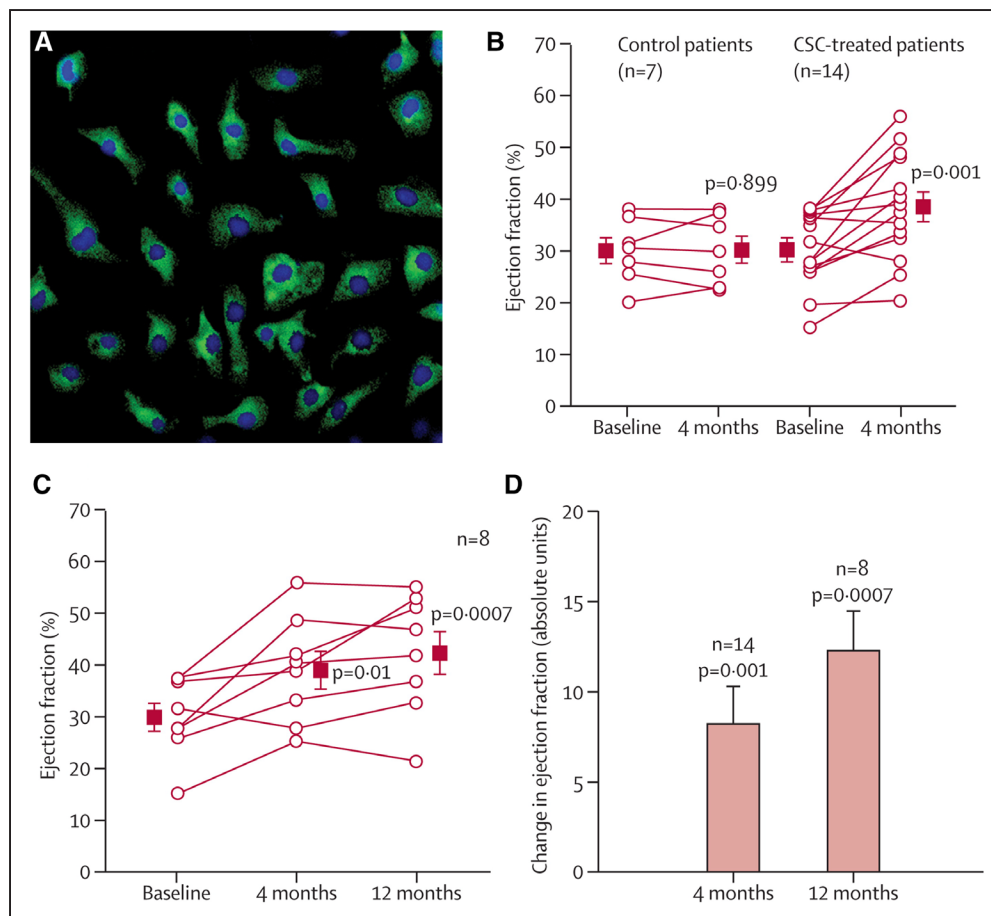


Figure 5. Results from the SCPIO trial. **A**, confocal image of autologous cardiac stem cells used in this study demonstrating c-kit positivity (green). Nuclei are stained blue. **B**, Left ventricular ejection fraction (LVEF; measured by echocardiography) at 4-month follow-up and baseline in control and treated patients. **C**, LVEF at 4 and 12 mo post-treatment in 8 cardiac stem cell (CSC)-treated patients. **D**, Absolute change in LVEF from baseline at 4 and 12 mo in the treated patients. Modified and reproduced from Bolli et al¹⁰⁵ with permission. Copyright ©2011, Elsevier.

cardiomyocytes *in vivo*, these cells confer salutary effects in preclinical models up to 1 year, supporting long-lasting paracrine signaling as an important mode of action.²⁹ CSCs were first tested in humans in the phase I, open label, randomized, controlled SCIPIO trial (Stem Cell Infusion in Patients With Ischemic Cardiomyopathy)^{105,117} in which CSCs were isolated from biopsies obtained during coronary artery bypass grafting in patients with ICM (LVEF $\leq 40\%$) and administered via intracoronary infusion 4 months later. These results were compared with control patients who underwent coronary artery bypass grafting but did not undergo any procedure to receive cells. In a per-protocol analysis of the initial results of 14 CSC-treated patients, LVEF measured by echocardiography increased by 8.2% compared with no change in 7 control patients. Furthermore, infarct size decreased by a mean of 9.8 g at 12-month follow-up among the patients who were able to undergo cMRI (n=9 CSC-treated patients; Figure 5). These encouraging findings are limited by lack of blinding and limited cMRI data, therefore the effect of CSCs is being further tested in ongoing, phase II CONCERT-HF trial (Combination of Mesenchymal and C-Kit+ Cardiac Stem Cells as Regenerative Therapy for Heart Failure),¹¹⁸ which will test transendocardial injection of MSCs, CSCs, and the combination of the 2 cell types in patients with ICM (NCT02501811).

Cardiosphere-Derived Cells

CDCs are a potentially promising mixture of cells, enriched for the CD105 cell surface receptor, resulting from culture outgrowths of heart biopsies that demonstrate increased paracrine activity compared with CSCs.⁴⁸ In the proof of concept CADUCEUS trial (Cardiosphere-Derived Autologous Stem Cells to Reverse Ventricular Dysfunction),¹⁰⁷ 33 patients were randomized to receive autologous CDCs or standard of care. In the 17 patients randomized to treatment, CDCs were grown from endomyocardial biopsies and delivered via intracoronary infusion 1.5 to 3 months after MI. CDC infusion produced a mean scar size reduction measured by cMRI of 7.7% at 6 months and 11.9% at 12 months compared with a nonsignificant increase in the control group by 0.3%. Reductions in scar size were accompanied by increases in viable myocardial mass. Despite the impact of CDC therapy on scar size, LVEDV, LVESV, or LVEF were not improved in treated patients compared with control patients. The effects of CDCs in ICM require further testing in larger efficacy trials. Allogeneic CDCs, which have a similar hypoimmunogenic profile to allogeneic MSCs and would eliminate the challenges of isolating and expanding each patient's own CDCs, were evaluated in the phase I/II ALLSTAR clinical trial (Allogeneic Heart Stem Cells to Achieve Myocardial Regeneration; NCT01458405) in patients with LV dysfunction secondary to an MI within the past 12 months. The primary efficacy outcome in this multicenter trial was the infarct size as assessed by cMRI.¹¹⁹ This trial was suspended by the Data Safety Monitoring Board because of evidence of futility to meet the primary end point.

Cardiopoeitic Stem Cells

Cardiopoeitic stem cells are second-generation therapeutic cells that include MSCs that have undergone treatment to optimize their cardioreparative functions.¹²⁰ The CHART-1 trial⁶

was a large, rigorous multicenter randomized, controlled trial assessing the efficacy of TESI of autologous, BM-MSCs treated with a specific cocktail to become cardiopoeitic stem cells. Cell administration was compared with sham procedure (LV catheterization and angiogram only) in patients with severe ICM (ejection fraction $<35\%$). The primary end point for this phase III trial was a hierarchical composite of a diverse array of end points with predefined thresholds, including mortality, worsening HF, MLHFQ score, 6MWD, LVESV, and LVEF (measured by echocardiography). At 39 weeks, the primary outcome showed no difference between sham and cell therapy. However, among patients with a larger baseline LVEDV ranging from 200 to 370 mL (60% of the patient population), cell treatment was associated with a significant improvement in the composite outcome, indicating that this subset of the population with more severe cardiac dilation may benefit from these cells. Another subanalysis revealed that patients with baseline LV volumes above the median did experience a significant improvement in the composite end point. Follow-up at 1 year, however, revealed a significant reduction in both LVESV and LVEDV (by 17 and 12.8 mL, respectively) in the cell-treated group over controls but no reduction in LVEF. Up to 21 injections of a defined concentration of cells were administered, depending on the availability of culture-expanded cardiopoeitic cells. Therefore, the number of injections is a proxy for dose. A subanalysis of the 1-year results revealed a trend but no significant differences, whereby patients who received a moderate number of injections (17–19) had improved reverse remodeling as compared with those who received >19 or <17 .¹²² Overall, this phase III study suggests that these cells impact remodeling albeit more convincingly in those patients with larger LV volumes at baseline, further highlighting the importance of selecting the optimal patient population for cell-based therapies.

Cell Combination Therapy

Cell combination therapy is another particularly promising new approach. This concept was derived from observations that MSC injections into porcine hearts upregulated the abundance of endogenous c-kit+ CSCs and cell cycle activity.²⁵ This suggested synergistic interactions between MSCs and CSCs in the cardiac repair process and led to a series of proof of concept studies in large animal models. These preclinical studies in a swine ICM model demonstrated enhanced cardioreparative effects of the combination of autologous CSCs and MSCs compared with MSCs alone. The combination product improved LVEF by $\approx 7\%$ compared with only 2.9% by MSCs alone.¹²³ A similar swine study using allogeneic CSCs and MSCs also demonstrated synergistic effects on cardiac repair and recovery without adverse immunologic events.⁷¹ These encouraging preclinical findings are being tested in the clinical setting in the CCTRN's CONCERT-HF trial (NCT02501811), a phase II randomized, placebo-controlled trial examining the safety and efficacy of NOGA catheter-based delivery of autologous MSCs, CSCs, and the combination of MSCs and CSCs.¹¹⁸

A specific fraction of autologous BMMNCs containing both MSCs and macrophages, Ixmyelocel-T, has also been tested in the setting of ICM. Patel et al¹²⁴ performed a multicenter, phase II trial assessing the efficacy of TESI of ixmyelocel-T (n=66) compared with placebo (n=60) in patients

with NYHA class III or IV ischemic HF. The composite of clinical cardiac events, including death, cardiac admissions, and emergency department visits, was significantly lower in the cell-treated group compared with placebo. Despite no significant differences in LV volumes, LVEF, or 6MWD, patients treated with ixmyelocel-T experienced a 37% reduction in cardiac events.¹²⁴ These findings are encouraging, especially given the use of a hard clinical end point rather than surrogate end points, and highlight the importance of seeking out novel cell combinations which confer an improved benefit over first-generation products.

Nonischemic Dilated Cardiomyopathy

Nonischemic dilated cardiomyopathy (NIDCM) is a severe and progressive syndrome in dire need of alternative treatments. Although less prevalent overall than ICM, NIDCM is now the leading diagnosis among adult heart transplant recipients.¹²⁵ In contrast to ICM, NIDCM is marked by a significant immunologic component, a lower scar burden, and a larger proportion of dysfunctional viable myocardium (DVM) that is a potential therapeutic target for cell-based therapies.¹²⁶ Using cMRI to measure the extent of scar and DVM, Bello et al¹²⁶ found that the amount of DVM is an independent predictor of improvements in LVEF, wall motion score, and LVESV index. Because stem and progenitor cell therapy works, at least in part, because of immunomodulatory signaling, reversal of myocardial dysfunction is a more attainable goal than replacement of scar, making NIDCM a prime setting for the application of cell-based therapies.

The clinical application of cell-based therapies for NIDCM is certainly at an early stage of development compared with AMI and ICM; however, the number of trials investigating stem cells in this population is increasing rapidly with promising results (Table 3). BMMNCs were the first cells tested in this population, with the TOPCARE-DCM

(Transplantation of Progenitor Cells and Recovery of Left Ventricular Function in Patients With Non Ischemic Dilative Cardiomyopathy),¹²⁷ ABCD (Autologous Bone Marrow Cells in Dilated Cardiomyopathy),¹²⁸ and MiHeart¹²⁹ trials, producing conflicting results. Trials of CD34+ cell treatment have noted more consistent improvements in LVEF.^{130,135} However, the pathogenesis of NIDCM in these trials varies widely, with some including only those with idiopathic pathogenesis,^{128,132} and others including mostly patients with viral¹³⁰ or Chagas-induced¹³⁶ NIDCM. Regardless of pathogenesis, a recent meta-analysis of randomized controlled trials of cell-based therapy in DCM found that bone marrow–derived cell therapy improved LVEF at 1 year by 3.53% but did not significantly change chamber volumes or patient functional capacity.²³ As opposed to ICM, in which cell-based therapy typically results in only modest improvements in LVEF and more marked changes in reverse remodeling, the greater impact on LVEF seen in NIDCM trials may be a result of the more abundant DVM. In both cardiomyopathies, inflammation plays an important role in the formation and maintenance of DVM, therefore stem cells with greater anti-inflammatory effects, such as MSCs, may be more beneficial for this population, as suggested by 2 recent trials using MSCs that reported marked improvements in LVEF.^{132,134}

The POSEIDON-DCM¹³² trial (Percutaneous Stem Cells Injection Delivery Effects on Neomyogenesis in Dilated Cardiomyopathy) is unique among trials for NIDCM because it compared the therapeutic effect of allogeneic and autologous BM-MSCs. Thirty-seven patients with idiopathic NIDCM were randomly allocated 1:1 to receive TESI of either allogeneic or autologous MSCs. At 12 months, LVEF improved from 27% to 35% in the allogeneic group but did not improve significantly in the autologous group (Figure 6). Interestingly, this striking improvement in LVEF was not accompanied by significant reductions in LV volumes suggesting that reverse remodeling is not

Table 3. Clinical Trials in Nonischemic Cardiomyopathy

Trial	Follow-Up	n	Cell Type and Dose	Delivery Method	End Point Evaluation Method	LVEF	LV Volumes	NYHA	Functional Capacity	QoL
TOPCARE-DCM Fischer-Rasokat et al ¹²⁷	1 y	33	BMMNC 259±135×10 ⁶	IC	LVG	↑	NS	N/A	N/A	N/A
ABCD Seth et al ¹²⁸	3 y	85	BMMNC 168±96×10 ⁶	IC	Echo	↑	↓ ESV	N/A	N/A	↑
MiHeart Martino et al ¹²⁹	12 mo	160	BMMNC Placebo	IC	Echo	NS	NS	NS	NS	NS
Vrtovec et al ^{130,131}	12 mo, 5 y	110	CD34+ from peripheral blood 113±26×10 ⁶ vs control	IC	Echo	↑	NS	N/A	↑	N/A
POSEIDON-DCM Hare et al ¹³²	12 m	37	1×10 ⁶ Allo BM-MSC 1×10 ⁶ Auto BM-MSC	TESI	cMRI/CT	↑	NS	NS	↑ Allo	↑ Allo
Butler et al ¹³³	3 mo (crossover)	22	Ischemia tolerant MSCs 1.5×10 ⁶ /kg vs placebo	IV	Echo/cMRI	↑	↑EDV	↑	↑	↑
Xiao et al ¹³⁴	12 mo	53	5.1×10 ⁸ BMMNC vs 4.9×10 ⁸ BM-MSCs vs placebo	IC	Echo	↑	↓ BMMNC	↑	NA	NA

↑ indicates increase; ↓, decrease; allo, allogeneic; auto, autologous; BMMNC, bone marrow mononuclear cell; cMRI, cardiac magnetic resonance imaging; CT, computed tomography; DCM, dilated cardiomyopathy; echo, echocardiography; EDV, end-diastolic volume; IC, intracoronary; IV, intravenous; LV, left ventricle; LVEF, left ventricular ejection fraction; LVG, left ventriculography; NA, not assessed; NS, not significant; NYHA, New York heart association; QoL, quality of life; and TESI, transendocardial stem cell injection.

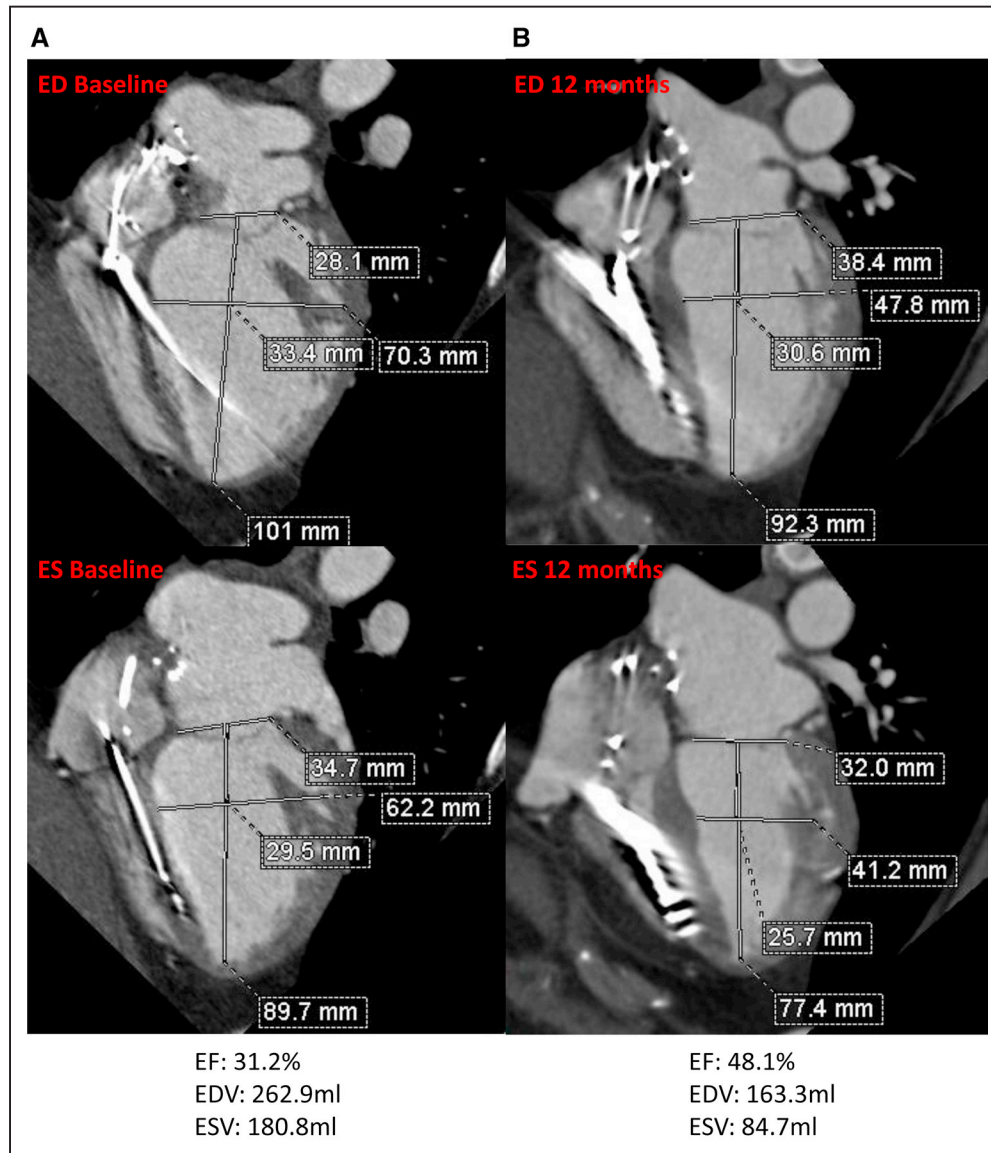


Figure 6. Ejection Fraction improvement in a patient from the POSEIDON-DCM trial. **A**, Cardiac computed tomography shows decreased global EF at baseline. **B**, Global EF has significantly improved at 12-months post TESI of MSCs. ED indicates end diastolic; EDV, end diastolic volume; ES, end systolic; ESV, end systolic volume; LVEF, left ventricular ejection fraction; MSC, mesenchymal stem cell; and TESI, Transendocardial Stem Cell Injection. Reprinted from Hare et al¹³² with permission. Copyright ©2017, Elsevier.

the primary means of cardiac functional improvement in this setting. Importantly, MSC treatment led to substantial and sustained declines in TNF- α levels that were reduced to a greater extent by allogeneic therapy than autologous. Furthermore, the 6MWD in the allogeneic group improved significantly more than in the autologous group, and the all-cause hospitalization rate and incidence of major adverse cardiac events were significantly lower in the allogeneic group.¹³² This study confirms the low immunogenicity of allogeneic MSCs and supports their superiority over autologous MSCs, likely because of reduced bioactivity among autologous cells because of aging and medical comorbidities. Although these results are exciting, this trial is limited by the lack of a control group and requires substantiation in a larger study.

In a pilot study by Butler et al,¹³³ BM-MSCs were isolated from healthy donors and grown under hypoxic conditions, which improves their migratory and secretory capacity, and delivered to 22 patients with NIDCM. Building on the

understanding that inflammation is a major driver of worsening HF and results in elevated systemic levels of inflammatory markers including TNF- α ,^{70,138} the authors hypothesized that intravenous administration would produce cardiac benefits for ICM patients. Although no improvements in LV anatomy or function were noted, 6MWD and QoL scores improved significantly in response to treatment even with the small sample size.¹³³ This study also noted changes in the percentage of immune cells, similar to the POSEIDON-DCM¹³² and TRIDENT⁶² trials and reductions in systemic TNF- α in response to cell treatment.

Lessons From Clinical Trials

The field has progressed substantially since the beginning of this century, and important lessons can be garnered from $\approx$ 2 decades of clinical research. At the same time, the series of disparate phase I and II trials currently available raise many

issues that require further clarification. First, to compare results between trials, there must be an effort to standardize measurement techniques, favoring those with higher reproducibility (such as cMRI). Second, the inclusion of multiple end points, including QoL and biomarkers, in phase II trials allows for greater insights into appropriate end point selection for phase III trials. Surrogate end points, such as LVEF and chamber volumes, are only markers associated with clinical outcomes such as mortality, therefore trials with longer follow-ups and analysis of rigorous clinical end points are important but lacking. Finally, phase III trials adequately powered to assess hard end points are needed to move the field forward.

Two major areas that require further investigation in clinical trials, and may explain the variation in clinical trial results to date, are mode of stem cell delivery and cell dose. Traditionally, these 2 factors are assessed early in preclinical and phase I trials of pharmaceuticals. Based on the biological nature and poorly understood modes of action of stem cells, determining optimal delivery and dosage remain particularly challenging. Because cell-based therapies use paracrine means as a major contributor to tissue repair, the number of cells delivered to and engrafting within the myocardium plays an important role in therapeutic efficacy. In a meta-analysis of preclinical studies, Kanelidis et al¹³⁹ found that TESI was associated with improved efficacy compared with intracoronary injection. Intracoronary injection requires stem cells to extravasate and migrate through the myocardium to reach areas of injury, which requires important homing signals that may be significantly reduced in chronic HF.¹³⁵ This concept is further supported by the observations of Vrtovc et al,¹³⁵ who performed one of the few clinical comparisons of cell delivery techniques by comparing intracoronary infusion and TESI of radiolabeled stem cells measured by single-photon emission computed tomography. They found higher cell retention was achieved by TESI which was associated with greater improvements in ventricular function, exercise capacity, and pro-B-type natriuretic peptide in patients with NIDCM.

Similar attention must be paid to cell dosing, which varies widely throughout trials. Thus far, dose selection has been rather arbitrary and based on convenience (ie, the maximum yield from bone marrow harvest). Importantly, some studies have shown that there can be a dosing threshold above which efficacy results are lost, highlighting the unique and challenging pharmacokinetics of biological agents.^{98,106,122} Furthermore, the potency of bone marrow-derived progenitor and stem cells is affected by patient characteristics, such as age and comorbidities, making direct comparisons of stem cell efficacy even more difficult.^{49,140} Although post hoc and meta-analyses have retrospectively examined the effect of cell dose,^{88,104,122} only a handful of trials were designed to specifically address dosing.^{45,62,84,141–143} These trials are extremely important guides for selecting the appropriate number of cells to administer, especially given that cellular retention via any delivery route is low and long-term cellular engraftment is even lower. The design of future trials assessing novel cell types should model dosing after carefully executed preclinical trials of large animals that include dose escalations.

Finally, the future of the clinical application of cell-based therapies in CRM relies on selection of the appropriate patient population. One of the most glaring inconsistencies across trials

is the variability in inclusion criteria and severity of cardiac and functional morbidity of the study populations. Post hoc analyses have suggested greater efficacy in younger patients,⁷ those with lower LVEF,¹⁴⁴ and those with larger LV volumes⁶ and are important to provide clues as to which patients are likely to respond to therapy. However, few trials have tackled this issue prospectively. Vrtovc et al¹⁴⁵ conducted a study of CD34⁺ cells for NIDCM in patients with normal glucose metabolism, insulin resistance, and diabetes mellitus and concluded that TESI of these cells is ineffective in patients with diabetes mellitus. Analysis of trial participants can provide genetic or molecular information to identify markers of therapeutic responders. For example, an analysis of the baseline biomarkers in responders and nonresponders to CD34⁺ therapy identified several immunologic markers that were associated with clinical response in NIDCM patients.¹⁴⁶ Another analysis of serum from a subset of patients from the CCTRN TIME and LateTIME trials identified several biomarkers that were significantly elevated in responders as compared with nonresponders.¹⁴⁷

Future Directions

The current data set assembled from cell-based therapy trials, albeit early stage, sets to stage for the design of a wave of new trials aimed at establishing clinical efficacy of the approach and adding refinements with regard to cell type, dosing, and delivery strategies. Among the key strategies that can be approached include the evaluation of potentially more potent cell types such as pluripotent cells, the relative efficacy of allogeneic cell preparations, and key issues around dosing and delivery, including multiple dosing regimens.

With regards to the use of human pluripotent stem cells, there is an amassing data set of preclinical studies that can form the basis for clinical investigation. ESCs are pluripotent cells derived from the inner cell mass of blastocysts and have been proposed as a potential therapy. In vitro, they can be differentiated toward a cardiac phenotype, which is thought to improve their survival and reparative potential. However, their use is limited because of ethical concerns, requirement for immunosuppression in the recipients, and the risk of tumor formation and arrhythmias. Although ESC-derived cardiomyocytes and cardiomyocyte progenitors have been tested in large animal models,¹⁴⁸ evidence of long-term engraftment is lacking.¹⁴⁹

The recently published ESCORT trial (Transplantation of Human Embryonic Stem Cell-Derived Progenitors in Severe Heart Failure)⁵³ is first human trial of ESC-derived cardiac progenitors. This small phase I study involved epicardial placement of a fibrin patch embedded with ESC cardiac progenitors during coronary artery bypass grafting. Of note, these patients experienced severe HF with a median LVEF of 26%. By 1 year, no patients experienced arrhythmias or had evidence of tumor formation. One patient died postoperatively from treatment-unrelated comorbidities and another died from HF at 22 months post-treatment. This study demonstrated the feasibility and short/medium-term safety of this treatment.⁵³ Whether these cells, delivered alone or in a fibrin patch, are truly efficacious warrants further clinical and preclinical investigation.

iPSCs, first discovered in 2006, are particularly exciting potential CRM therapeutics because they can be reprogrammed to a pluripotent state similar to ESCs but do not harbor similar

ethical and availability concerns.¹⁵⁰ These cells are posited to function similarly to other stem cell therapeutics by contributing to cardiac function via remuscularization and trophic effects. These cells are currently undergoing rapid translation into large animal models, and their use in human trials is on the horizon. Kawamura et al¹⁵¹ implanted sheets of human iPSC-derived cardiomyocytes, endothelial cells, and smooth muscle cells in a pig model of MI and noted improved cardiac function post-treatment. Similar to ESCs, the translation of iPSCs into clinical investigation as a therapy for CVD will require further research into how to form mature cardiomyocytes from these cells that can electromechanically couple with resident cardiomyocytes and not form tumors.

In terms of techniques to optimize cell products and delivery methods, a key consideration is the use of bioengineering techniques to create scaffolds for cell products to be delivered as patches or suspended in hydrogels with the goal of improving cell retention and survival in host tissue.¹⁵² Another optimization method under investigation is conditioning of the target myocardium to enhance reception and efficacy of transplanted cells. This technique was examined in the CELLWAVE trial (Combined Extracorporeal Shock Wave Therapy and Intracoronary Cell Therapy in Chronic Ischemic Myocardium)¹⁵³ in which extracorporeal shockwaves were delivered to the myocardium before intracoronary delivery of BMMNCs. At 4 months, the shock wave+BMMNC group experienced an improvement in absolute ejection fraction of 3.2% compared with no improvement in the shock wave+placebo or BMMNC-alone groups. This therapy requires testing in larger clinical trials and may achieve maximal benefits when combined with more efficacious or optimized cell types.

Modifications or pretreatment of the cells themselves is also a promising option to modify their efficacy, as with cardiopoietic cells. Gene overexpression is one means to enhance the therapeutic efficacy and survival of stem cells. To overcome issues with poor mechanical coupling with the host myocardium, Gwizdala et al¹⁵⁴ transfected skeletal myoblasts with the gene for the gap junction protein connexin-43. These cells were then delivered to 13 patients with severe HF by TESI, and by 6 months patients in this pilot study experience improvements in NYHA class and exercise capacity, with a trend toward improvement in LV chamber dimensions. Most importantly, none of the patients experienced ventricular arrhythmias, one of the major concerns with using skeletal myoblasts because they naturally have poor expression of connexin-43.¹⁵⁴ Another randomized, placebo-controlled trial will test intracoronary infusion of autologous circulating endothelial progenitor cells grown in culture for 3 days and transfected with endothelial nitric oxide synthase.^{154a} This trial will be one of the largest to date to examine genetically modified progenitor cells for CVD.

Other cellular modifications involving preconditioning of cells in vitro have been investigated in preclinical studies and may soon be translated to the clinical arena. Hypoxic preconditioning or growth of cells in a low oxygen environment improves their survival, resistance to oxidative stress, and migratory capacity.¹⁵⁵ Hypoxia conditioned or ischemia tolerant MSCs were used in a phase II trial testing intravenous administration in patients with NIDCM.¹⁵⁶

The hypothesis that stem cells must engraft in the target tissue long-term to either directly remuscularize or stimulate repair via local paracrine signaling is currently being challenged. In fact, intravenous administration of stem cells, once thought to be ineffective because of resultant low cardiac engraftment rates, has been investigated in AMI⁹⁴ and NIDCM¹³³ and suggest that systemic and cardiac improvements may result from intravenous therapy. Because paracrine signaling is the primary means of stem cell efficacy, the future of cell-based therapies for CVD may include treatment with cell-free suspensions of paracrine molecules produced by cells in vivo, including exosomes derived from stem cells, which carry mRNA, miRNAs, and other signaling molecules that can influence and reprogram host cells.¹⁵⁷ The similar efficacy of exosomes compared with their parent cells has been demonstrated in iPSCs¹⁵⁸ and ESCs,¹⁵⁹ which could potentially provide benefits without the tumorigenic risk associated with pluripotent cells. Some groups are even working to create completely cell-free microparticles containing molecules derived from stem cells that can release these factors in a controlled manner.¹⁶⁰ Such cell mimicking microparticles and other cell-free products derived from stem cells may be a solution to the barriers of working with live cells.

Finally, most pharmaceutical therapies in existence are delivered multiple times to achieve efficacy, and it may be that stem cells are no exception. Preclinical evidence from Tokita et al¹⁶¹ indicate that repeated injections over time improve the clinical efficacy of cardiac progenitor cells. Similarly, Guo et al¹⁶² demonstrated that 3 doses of cardiac-derived MSCs yield greater improvements in cardiac function in a preclinical MI model despite minimal engraftment. Repeated injections may be the paradigm shift needed in cell-based therapies to achieve long-term and meaningful clinical improvements. The impact of repeated TESI of peripheral CD34+ cells in patients with dilated cardiomyopathy is currently being evaluated in the ongoing REMEDIUM trial (Repetitive Intramyocardial CD34+ Cell Therapy in Dilated Cardiomyopathy; NCT022248532). Similarly, single versus repeated intracoronary infusion of BMMNCs in patients with chronic ICM is being examined in the REPEAT trial (Compare the Effects of Single Versus Repeated Intracoronary Application of Autologous Bone Marrow-Derived Mononuclear Cells on Mortality in Patients With Chronic Post-Infarction Heart Failure; NCT01693042).

Conclusions

Nearly 2 decades of research supports the idea that cellular therapy to promote cardiac regeneration and repair is safe and potentially promising. To date, no trial has been conducted with sufficient statistical power to demonstrate efficacy of cell therapy to reduce mortality. Nevertheless, multiple cell types have been investigated for a variety of cardiac conditions, and important lessons about trial design have been learned and are guiding the first wave of ongoing phase III trials. As technologies improve, engineered and modified cell types along with rationally designed cell combination therapy can be submitted to early stage proof of concept clinical testing. Modified adult stem cells and human pluripotent cells will likely be tested imminently, as

will cell-free products, such as stem cell–derived exosomes and microparticles. Finally, the use of engineered scaffolds and repeated administrations of cells will be tested soon and will move us to the next phase of the field. At a minimum, the field can expect the results of clinically powered phase III trials by the end of this decade.

Acknowledgments

We thank Wayne Balkan, PhD, for his assistance in critical review and editing.

Sources of Funding

This work was funded by National Institutes of Health (NIH) grant 5UM HL113460 to J.M. Hare. J.M. Hare is also funded by NIH grants R01 HL107110 and R01HL137355, the Starr Foundation, and the Soffer Family Foundation.

Disclosures

J.M. Hare reports having a patent for cardiac cell-based therapy and holds equity in Vestion Inc and maintains a professional relationship with Vestion Inc as a consultant and member of the Board of Directors and Scientific Advisory Board. Vestion Inc did not play a role in the design, conduct, or funding of the study. J.M. Hare is the Chief Scientific Officer, a compensated consultant, and advisory board member for Longeveron and holds equity in Longeveron. J.M. Hare is also the coinventor of intellectual property licensed to Longeveron. The other authors report no conflicts.

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